

- Q.22 Runaway reactions
- Q.23 Dow's Fire and Explosion Index (FEI)
- Q.24 GHS
- Q.25 Process safety management
- Q.26 LC 50
- Q.27 Onsite Plan
- Q.28 Non-destructive testing
- Q.29 Event Tree Analysis
- Q.30 Incompatible chemicals
- Q.31 Colour and number coding of chemicals
- Q.32 HAZOP
- Q.33 Process Hazards and its Inspection Techniques
- Q.34 Safety valve
- Q.35 Process Safety Information (PSI)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Describe the concept of a runaway reaction in the chemical process industry. Discuss the factors that can lead to a runaway reaction, including process parameters, equipment failures, and human error. Provide examples of real-world incidents where runaway reactions have occurred and discuss the consequences.
- Q.37 Discuss the role of simulation and modeling in enhancing emergency preparedness. Provide examples of how simulation and modeling tools are utilized to assess and mitigate risks, develop response plans, and improve resilience in the face of various emergencies and disasters.
- Q.38 Unconfined vapor cloud explosions (UVCEs) are a significant hazard in industrial environments, posing risks to personnel, facilities, and surrounding communities. Discuss the characteristics, causes, consequences, and prevention/mitigation strategies of UVCEs in industrial settings, providing examples and emphasizing the importance of safety measures and regulatory compliance.

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Roll No.

1st Year / Industrial Safety

Subject:- Safety in Chemical Industry

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following models is commonly used to represent the dispersion of pollutants in the atmosphere?
 - a) Gaussian model
 - b) Exponential model
 - c) Linear model
 - d) Logarithmic model
- Q.2 What is the primary mechanism by which static electricity is generated when combing hair?
 - a) Conduction of electrons between the comb and hair
 - b) Induction between the comb and hair
 - c) Convection of charge from the hair
 - d) Radiation from the comb to the hair
- Q.3 What does MSDS stand for in the context of chemical safety?
 - a) Material Safety data Sheet
 - b) Multi-stage Distillation System
 - c) Molecular Structure Determination System
 - d) Mass Spectrometry Data Sheet
- Q.4 Which of the following is not considered a chemical-related health hazard?
 - a) Irritancy
 - b) Corrosivity
 - c) Flammability
 - d) Radiation
- Q.5 When transferring chemicals from a labeled container to a portable one, under what circumstances are you exempt from standard hazardous material labeling requirements?
 - a) When you hand the container off to someone else
 - b) When you leave the work area before using the materials
 - c) When you don't use the materials before the end of your work shift
 - d) None of the above

- Q.6 At what concentration does ammonia pose an immediate danger to life and health?
 a) 50 ppm b) 100 ppm
 c) 500 ppm d) 1000 ppm
- Q.7 Which law in India mandates the provision of MSDS (Material Safety Data Sheet)?
 a) The Environment Protection Act, 1986
 b) The chemical Weapons convention Act, 2000
 c) The hazardous Waste Management Rules, 2016
 d) The manufacture, Storage and Import of Hazardous Chemicals Rules, 1989
- Q.8 What does GHS stand for in chemical safety regulations?
 a) Global harmonized standard
 b) General health and safety
 c) Globally Harmonized System
 d) General Hazardous Substances
- Q.9 What does TLV (Threshold Limit Value) represent in occupational health and safety?
 a) The maximum allowable concentration of a substance in workplace air
 b) The minimum concentration of a substance required for its detection
 c) The level of toxicity of a substance in a laboratory setting
 d) The minimum quantity of a substance that triggers an alarm system
- Q.10 Through which route (s) can hazardous chemicals enter the human body?
 a) Inhalation b) Absorption
 c) Ingestion d) All of the above

SECTION-B

Note: Objective type questions. All questions are compulsory.
 (10x1=10)

- Q.11 The flash point of a substance is the _____ at which it can vaporize to form an ignitable mixture in air near the surface of the liquid.

(2)

RLI-095407

- Q.12 Fire and Explosion Index Rating (FEIR), is a measure used in _____ safety management to assess the risk of fire and explosion associated with the handling, storage, and processing of hazardous materials in industrial facilities.
- Q.13 A the UN number on a hazardous material label indicate the hazard class of the material.
 a) True b) False
- Q.14 Lethal Concentration 50 and is a measure used in _____ to assess the concentration of a substance in air or water that is lethal to 50% of the population of test organisms within a specified exposure period.
- Q.15 _____ instrumentation is used to measure the rate of fluid flow in pipelines.
- Q.16 Stress testing is not a common non-destructive testing (NDT) technique used in chemical plant inspections?
 a) True b) False
- Q.17 The full form for Acronym “API” stand for in the context of inspection standards for process equipment is _____?
- Q.18 Atmospheric pressure cannot affect the Upper Explosive Limit (UEL) of a substance?
 a) True b) False
- Q.19 Which organization provides codes and standards for the design, fabrication, and inspection of pressure vessels?
- Q.20 The purpose of Failure Mode and Effects Analysis (FMEA) in process reliability is to identify potential failure modes and their effects on system performance.
 a) True b) False

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions.
 (12x5=60)

- Q.21 Failure Mode and Effects Analysis (FEMA) in pressure vessel safety.

(3)

RLI-095407